

Algebraic Topology: a beginner's course

This lecture:

**AlgTop29: Universal
covering spaces**

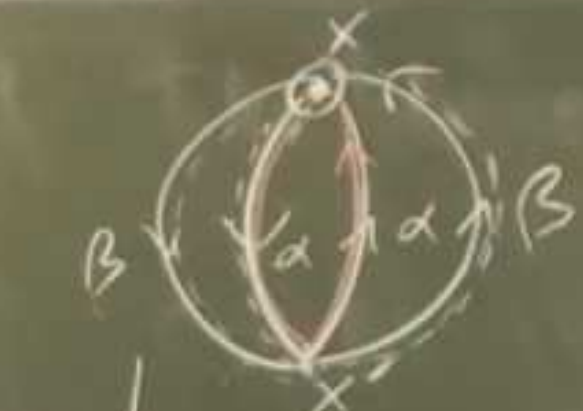
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Lecture 29 The Universal covering space



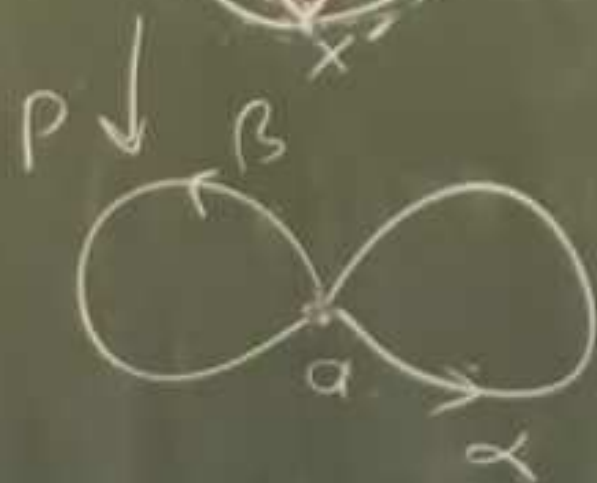
Ex. 3

X



$\pi(X, x) : \alpha^2, \beta^2, \alpha\beta$

B



$\pi(B, a) : \text{words in } \alpha, \beta :$

eg. $\alpha^3 \beta^2 \alpha^{-5} \beta^7 \alpha'$

Idea: Covering spaces of $B \Leftrightarrow$ subgroups of $\pi(B)$

The smallest subgp of $\pi(B) \hookrightarrow e \Leftrightarrow$ the biggest covering space of B
The universal cover

Eg.



$A = \tilde{X}$ univ. cover.

idea: identify $B = S^1$
 $z \sim [\alpha]$



α path in B starting at a .

lift α to X , get path from x to z

z ind. of representative of $[\alpha]$

the universal cover of B :

then define the points of

classes of paths in B

α : path in B starting at

by

point of α (point in

Ex

